In [3]:
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import numpy as np
import itertools
import matplotlib.pyplot as plt

# ------------------------------
# STEP 1: Define problem constants
# ------------------------------

days = 5  # Total number of days
raw_material_max = 10  # Max raw material units per day
market_states = [1, 2]  # 1: High demand A, 2: High demand B

# Define available actions and corresponding units of A and B produced
actions = {
    "Produce_2A_0B": (2, 0),  # Uses 4 RM
    "Produce_1A_2B": (1, 2),  # Uses 4 RM
    "Produce_0A_5B": (0, 5),  # Uses 5 RM
    "Produce_3A_0B": (3, 0),  # Uses 6 RM
    "Do_Nothing": (0, 0)      # Uses 0 RM
}

gamma = 1.0       # Discount factor (set to 1 as horizon is short)
theta = 0.001     # Threshold for convergence

# ------------------------------------------------
# STEP 2: Define reward function and valid actions
# ------------------------------------------------

# Computes reward based on units of A and B produced and the market state
def get_reward(a_count, b_count, market_state):
    if market_state == 1:
        return a_count * 8 + b_count * 2
    else:
        return a_count * 3 + b_count * 5

# Returns only those actions that are feasible within the available RM
def valid_actions(rm):
    valid = []
    for action, (a, b) in actions.items():
        cost = a * 2 + b * 1  # A costs 2 RM, B costs 1 RM
        if cost <= rm:
            valid.append((action, a, b, cost))
    return valid

# ------------------------------------------------
# STEP 3: Initialize state-value function and policy
# ------------------------------------------------

# State: (day, market_state, remaining_raw_material)
states = list(itertools.product(range(1, days + 1), market_states, range(0, raw_

V = {}        # State-value function
policy = {}   # Optimal policy per state
```

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for s in states:
    V[s] = 0
    policy[s] = "Do_Nothing"

# ------------------------------------------------------
# STEP 4: Value Iteration algorithm with iteration print
# ------------------------------------------------------

iteration = 0
delta_history = []  # To track convergence for plotting

while True:
    delta = 0  # Max value change in this iteration
    print(f"\n----- Iteration {iteration + 1} -----")

    for state in states:
        day, market, rm = state

        # Skip if beyond final day
        if day > days:
            continue

        best_value = float('-inf')
        best_action = "Do_Nothing"

        # Evaluate all valid actions
        for action_name, a, b, cost in valid_actions(rm):
            reward = get_reward(a, b, market)
            next_day = day + 1

            # If final day, no future value
            if next_day > days:
                next_val = 0
            else:
                # Next state: average over possible market states (each 50%)
                next_val = 0.5 * V.get((next_day, 1, raw_material_max), 0) + \
                           0.5 * V.get((next_day, 2, raw_material_max), 0)

            total_value = reward + gamma * next_val

            # Select best action
            if total_value > best_value:
                best_value = total_value
                best_action = action_name

        # Update value function and policy
        delta = max(delta, abs(V[state] - best_value))
        V[state] = best_value
        policy[state] = best_action
        print(f"State={state} -> BestAction={best_action}, Value={round(best_val

    delta_history.append(delta)
    iteration += 1

    # Convergence check
    if delta < theta:
        print("\nConverged.")
        break
```

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# ------------------------------------------
# STEP 5: Plot value convergence over iterations
# ------------------------------------------

plt.figure(figsize=(10, 5))
plt.plot(range(1, len(delta_history)+1), delta_history, marker='o')
plt.xlabel("Iteration")
plt.ylabel("Max Value Function Change (Δ)")
plt.title("Value Convergence over Iterations")
plt.grid(True)
plt.show()

# ------------------------------------------------
# STEP 6: Display optimal policy for Day 1 with 10 RM
# ------------------------------------------------

print("\nOptimal Actions on Day 1 with 10 RM:")
for market in market_states:
    s = (1, market, raw_material_max)
    print(f"Market State {market}: Action -> {policy[s]}")

# ------------------------------------------------
# STEP 7: Simulate the learned policy for 1000 runs
# ------------------------------------------------

def simulate_policy(runs=1000):
    total_profit = 0
    for _ in range(runs):
        profit = 0
        for d in range(1, days + 1):
            market = np.random.choice([1, 2])  # Random market
            state = (d, market, raw_material_max)
            action = policy[state]
            a, b = actions[action]
            reward = get_reward(a, b, market)
            profit += reward
        total_profit += profit
    return total_profit / runs

avg_profit = simulate_policy()
print(f"\nAverage Total Profit over 1000 Simulations: ${round(avg_profit, 2)}")
```

```
----- Iteration 1 -----
State=(1, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(1, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(1, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(1, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(1, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(1, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(1, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(1, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(1, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(1, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(1, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(1, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(1, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(1, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(1, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(1, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(1, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(1, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(1, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(1, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(1, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(1, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0
State=(2, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(2, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(2, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(2, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(2, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(2, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(2, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(2, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(2, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(2, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(2, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(2, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(2, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(2, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(2, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(2, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(2, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(2, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(2, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(2, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(2, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(2, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0
State=(3, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(3, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(3, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(3, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(3, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(3, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(3, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(3, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(3, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(3, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(3, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(3, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(3, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(3, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(3, 2, 3) -> BestAction=Do_Nothing, Value=0.0
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State=(3, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(3, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(3, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(3, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(3, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(3, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(3, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0
State=(4, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(4, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(4, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(4, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(4, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(4, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(4, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(4, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(4, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(4, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(4, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(4, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(4, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(4, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(4, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(4, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(4, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(4, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(4, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(4, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(4, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(4, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(5, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0

----- Iteration 2 -----
State=(1, 1, 0) -> BestAction=Do_Nothing, Value=24.5
State=(1, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(1, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(1, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(1, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(1, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(1, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
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State=(1, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(1, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
State=(1, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(1, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(1, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(1, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(1, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(1, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(1, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(1, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
State=(1, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(1, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(1, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(1, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(1, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(2, 1, 0) -> BestAction=Do_Nothing, Value=24.5
State=(2, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(2, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(2, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(2, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(2, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(2, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
State=(2, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(2, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
State=(2, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(2, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(2, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(2, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(2, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(2, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(2, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(2, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
State=(2, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(2, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(2, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(2, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(2, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(3, 1, 0) -> BestAction=Do_Nothing, Value=24.5
State=(3, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(3, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(3, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(3, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(3, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(3, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
State=(3, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(3, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
State=(3, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(3, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(3, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(3, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(3, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(3, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(3, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(3, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
State=(3, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(3, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(3, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(3, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(3, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 1, 0) -> BestAction=Do_Nothing, Value=24.5
```

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State=(4, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(4, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(5, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(5, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0

----- Iteration 3 -----
State=(1, 1, 0) -> BestAction=Do_Nothing, Value=49.0
State=(1, 1, 1) -> BestAction=Do_Nothing, Value=49.0
State=(1, 1, 2) -> BestAction=Do_Nothing, Value=49.0
State=(1, 1, 3) -> BestAction=Do_Nothing, Value=49.0
State=(1, 1, 4) -> BestAction=Produce_2A_0B, Value=65.0
State=(1, 1, 5) -> BestAction=Produce_2A_0B, Value=65.0
State=(1, 1, 6) -> BestAction=Produce_3A_0B, Value=73.0
State=(1, 1, 7) -> BestAction=Produce_3A_0B, Value=73.0
State=(1, 1, 8) -> BestAction=Produce_3A_0B, Value=73.0
State=(1, 1, 9) -> BestAction=Produce_3A_0B, Value=73.0
State=(1, 1, 10) -> BestAction=Produce_3A_0B, Value=73.0
State=(1, 2, 0) -> BestAction=Do_Nothing, Value=49.0
State=(1, 2, 1) -> BestAction=Do_Nothing, Value=49.0
State=(1, 2, 2) -> BestAction=Do_Nothing, Value=49.0
State=(1, 2, 3) -> BestAction=Do_Nothing, Value=49.0
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State=(1, 2, 4) -> BestAction=Produce_1A_2B, Value=62.0
State=(1, 2, 5) -> BestAction=Produce_0A_5B, Value=74.0
State=(1, 2, 6) -> BestAction=Produce_0A_5B, Value=74.0
State=(1, 2, 7) -> BestAction=Produce_0A_5B, Value=74.0
State=(1, 2, 8) -> BestAction=Produce_0A_5B, Value=74.0
State=(1, 2, 9) -> BestAction=Produce_0A_5B, Value=74.0
State=(1, 2, 10) -> BestAction=Produce_0A_5B, Value=74.0
State=(2, 1, 0) -> BestAction=Do_Nothing, Value=49.0
State=(2, 1, 1) -> BestAction=Do_Nothing, Value=49.0
State=(2, 1, 2) -> BestAction=Do_Nothing, Value=49.0
State=(2, 1, 3) -> BestAction=Do_Nothing, Value=49.0
State=(2, 1, 4) -> BestAction=Produce_2A_0B, Value=65.0
State=(2, 1, 5) -> BestAction=Produce_2A_0B, Value=65.0
State=(2, 1, 6) -> BestAction=Produce_3A_0B, Value=73.0
State=(2, 1, 7) -> BestAction=Produce_3A_0B, Value=73.0
State=(2, 1, 8) -> BestAction=Produce_3A_0B, Value=73.0
State=(2, 1, 9) -> BestAction=Produce_3A_0B, Value=73.0
State=(2, 1, 10) -> BestAction=Produce_3A_0B, Value=73.0
State=(2, 2, 0) -> BestAction=Do_Nothing, Value=49.0
State=(2, 2, 1) -> BestAction=Do_Nothing, Value=49.0
State=(2, 2, 2) -> BestAction=Do_Nothing, Value=49.0
State=(2, 2, 3) -> BestAction=Do_Nothing, Value=49.0
State=(2, 2, 4) -> BestAction=Produce_1A_2B, Value=62.0
State=(2, 2, 5) -> BestAction=Produce_0A_5B, Value=74.0
State=(2, 2, 6) -> BestAction=Produce_0A_5B, Value=74.0
State=(2, 2, 7) -> BestAction=Produce_0A_5B, Value=74.0
State=(2, 2, 8) -> BestAction=Produce_0A_5B, Value=74.0
State=(2, 2, 9) -> BestAction=Produce_0A_5B, Value=74.0
State=(2, 2, 10) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 1, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 4) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 5) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 6) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 7) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 8) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 9) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 10) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 2, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 4) -> BestAction=Produce_1A_2B, Value=62.0
State=(3, 2, 5) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 6) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 7) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 8) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 9) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 10) -> BestAction=Produce_0A_5B, Value=74.0
State=(4, 1, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
```

```
State=(4, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(4, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(5, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(5, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0

----- Iteration 4 -----
State=(1, 1, 0) -> BestAction=Do_Nothing, Value=73.5
State=(1, 1, 1) -> BestAction=Do_Nothing, Value=73.5
State=(1, 1, 2) -> BestAction=Do_Nothing, Value=73.5
State=(1, 1, 3) -> BestAction=Do_Nothing, Value=73.5
State=(1, 1, 4) -> BestAction=Produce_2A_0B, Value=89.5
State=(1, 1, 5) -> BestAction=Produce_2A_0B, Value=89.5
State=(1, 1, 6) -> BestAction=Produce_3A_0B, Value=97.5
State=(1, 1, 7) -> BestAction=Produce_3A_0B, Value=97.5
State=(1, 1, 8) -> BestAction=Produce_3A_0B, Value=97.5
State=(1, 1, 9) -> BestAction=Produce_3A_0B, Value=97.5
State=(1, 1, 10) -> BestAction=Produce_3A_0B, Value=97.5
State=(1, 2, 0) -> BestAction=Do_Nothing, Value=73.5
State=(1, 2, 1) -> BestAction=Do_Nothing, Value=73.5
State=(1, 2, 2) -> BestAction=Do_Nothing, Value=73.5
State=(1, 2, 3) -> BestAction=Do_Nothing, Value=73.5
State=(1, 2, 4) -> BestAction=Produce_1A_2B, Value=86.5
State=(1, 2, 5) -> BestAction=Produce_0A_5B, Value=98.5
State=(1, 2, 6) -> BestAction=Produce_0A_5B, Value=98.5
State=(1, 2, 7) -> BestAction=Produce_0A_5B, Value=98.5
State=(1, 2, 8) -> BestAction=Produce_0A_5B, Value=98.5
State=(1, 2, 9) -> BestAction=Produce_0A_5B, Value=98.5
State=(1, 2, 10) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 1, 0) -> BestAction=Do_Nothing, Value=73.5
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State=(2, 1, 1) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 2) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 3) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 4) -> BestAction=Produce_2A_0B, Value=89.5
State=(2, 1, 5) -> BestAction=Produce_2A_0B, Value=89.5
State=(2, 1, 6) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 7) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 8) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 9) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 10) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 2, 0) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 1) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 2) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 3) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 4) -> BestAction=Produce_1A_2B, Value=86.5
State=(2, 2, 5) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 6) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 7) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 8) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 9) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 10) -> BestAction=Produce_0A_5B, Value=98.5
State=(3, 1, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 4) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 5) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 6) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 7) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 8) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 9) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 10) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 2, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 4) -> BestAction=Produce_1A_2B, Value=62.0
State=(3, 2, 5) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 6) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 7) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 8) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 9) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 10) -> BestAction=Produce_0A_5B, Value=74.0
State=(4, 1, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(4, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
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State=(4, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(5, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(5, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0

----- Iteration 5 -----
State=(1, 1, 0) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 1) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 2) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 3) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 4) -> BestAction=Produce_2A_0B, Value=114.0
State=(1, 1, 5) -> BestAction=Produce_2A_0B, Value=114.0
State=(1, 1, 6) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 7) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 8) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 9) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 10) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 2, 0) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 1) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 2) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 3) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 4) -> BestAction=Produce_1A_2B, Value=111.0
State=(1, 2, 5) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 6) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 7) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 8) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 9) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 10) -> BestAction=Produce_0A_5B, Value=123.0
State=(2, 1, 0) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 1) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 2) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 3) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 4) -> BestAction=Produce_2A_0B, Value=89.5
State=(2, 1, 5) -> BestAction=Produce_2A_0B, Value=89.5
State=(2, 1, 6) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 7) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 8) -> BestAction=Produce_3A_0B, Value=97.5
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State=(2, 1, 9) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 10) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 2, 0) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 1) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 2) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 3) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 4) -> BestAction=Produce_1A_2B, Value=86.5
State=(2, 2, 5) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 6) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 7) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 8) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 9) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 10) -> BestAction=Produce_0A_5B, Value=98.5
State=(3, 1, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 4) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 5) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 6) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 7) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 8) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 9) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 10) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 2, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 4) -> BestAction=Produce_1A_2B, Value=62.0
State=(3, 2, 5) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 6) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 7) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 8) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 9) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 10) -> BestAction=Produce_0A_5B, Value=74.0
State=(4, 1, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(4, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(5, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 2) -> BestAction=Do_Nothing, Value=0.0
```

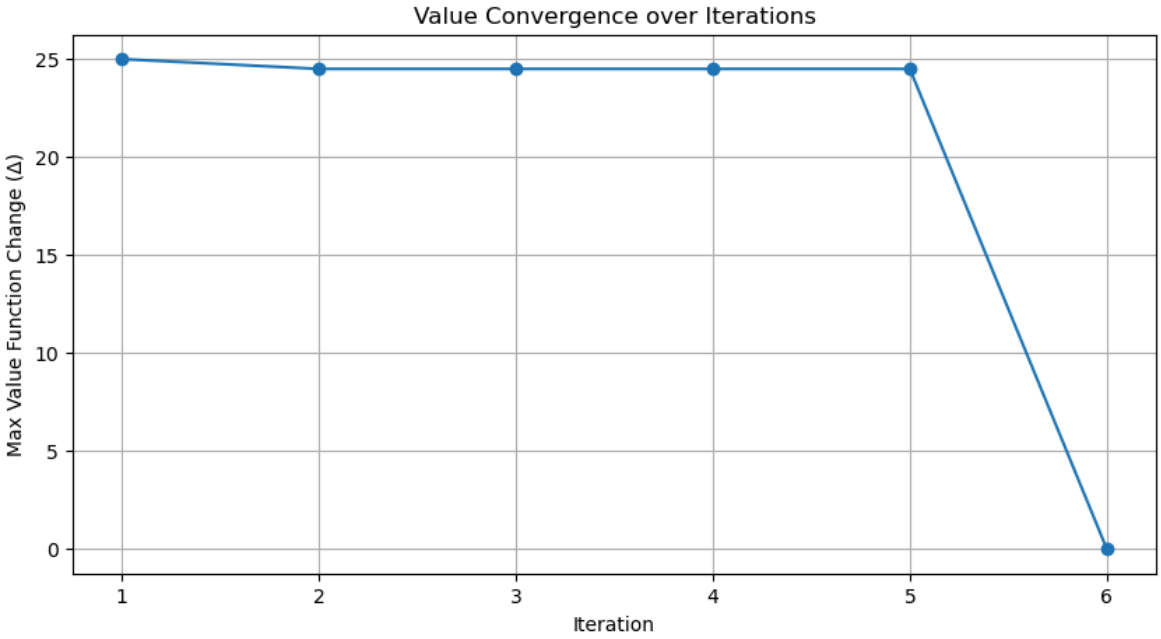
```
State=(5, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(5, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0

----- Iteration 6 -----
State=(1, 1, 0) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 1) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 2) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 3) -> BestAction=Do_Nothing, Value=98.0
State=(1, 1, 4) -> BestAction=Produce_2A_0B, Value=114.0
State=(1, 1, 5) -> BestAction=Produce_2A_0B, Value=114.0
State=(1, 1, 6) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 7) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 8) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 9) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 1, 10) -> BestAction=Produce_3A_0B, Value=122.0
State=(1, 2, 0) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 1) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 2) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 3) -> BestAction=Do_Nothing, Value=98.0
State=(1, 2, 4) -> BestAction=Produce_1A_2B, Value=111.0
State=(1, 2, 5) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 6) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 7) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 8) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 9) -> BestAction=Produce_0A_5B, Value=123.0
State=(1, 2, 10) -> BestAction=Produce_0A_5B, Value=123.0
State=(2, 1, 0) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 1) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 2) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 3) -> BestAction=Do_Nothing, Value=73.5
State=(2, 1, 4) -> BestAction=Produce_2A_0B, Value=89.5
State=(2, 1, 5) -> BestAction=Produce_2A_0B, Value=89.5
State=(2, 1, 6) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 7) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 8) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 9) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 1, 10) -> BestAction=Produce_3A_0B, Value=97.5
State=(2, 2, 0) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 1) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 2) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 3) -> BestAction=Do_Nothing, Value=73.5
State=(2, 2, 4) -> BestAction=Produce_1A_2B, Value=86.5
State=(2, 2, 5) -> BestAction=Produce_0A_5B, Value=98.5
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State=(2, 2, 6) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 7) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 8) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 9) -> BestAction=Produce_0A_5B, Value=98.5
State=(2, 2, 10) -> BestAction=Produce_0A_5B, Value=98.5
State=(3, 1, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 1, 4) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 5) -> BestAction=Produce_2A_0B, Value=65.0
State=(3, 1, 6) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 7) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 8) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 9) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 1, 10) -> BestAction=Produce_3A_0B, Value=73.0
State=(3, 2, 0) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 1) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 2) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 3) -> BestAction=Do_Nothing, Value=49.0
State=(3, 2, 4) -> BestAction=Produce_1A_2B, Value=62.0
State=(3, 2, 5) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 6) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 7) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 8) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 9) -> BestAction=Produce_0A_5B, Value=74.0
State=(3, 2, 10) -> BestAction=Produce_0A_5B, Value=74.0
State=(4, 1, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 1, 4) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 5) -> BestAction=Produce_2A_0B, Value=40.5
State=(4, 1, 6) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 7) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 8) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 9) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 1, 10) -> BestAction=Produce_3A_0B, Value=48.5
State=(4, 2, 0) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 1) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 2) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 3) -> BestAction=Do_Nothing, Value=24.5
State=(4, 2, 4) -> BestAction=Produce_1A_2B, Value=37.5
State=(4, 2, 5) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 6) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 7) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 8) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 9) -> BestAction=Produce_0A_5B, Value=49.5
State=(4, 2, 10) -> BestAction=Produce_0A_5B, Value=49.5
State=(5, 1, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 1, 4) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 5) -> BestAction=Produce_2A_0B, Value=16.0
State=(5, 1, 6) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 7) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 8) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 9) -> BestAction=Produce_3A_0B, Value=24.0
State=(5, 1, 10) -> BestAction=Produce_3A_0B, Value=24.0
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State=(5, 2, 0) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 1) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 2) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 3) -> BestAction=Do_Nothing, Value=0.0
State=(5, 2, 4) -> BestAction=Produce_1A_2B, Value=13.0
State=(5, 2, 5) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 6) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 7) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 8) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 9) -> BestAction=Produce_0A_5B, Value=25.0
State=(5, 2, 10) -> BestAction=Produce_0A_5B, Value=25.0

Converged.
```




```
Optimal Actions on Day 1 with 10 RM:
Market State 1: Action -> Produce_3A_0B
Market State 2: Action -> Produce_0A_5B

Average Total Profit over 1000 Simulations: $122.5
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In [ ]:

Conclusion

In this assignment, we implemented a dynamic programming-based simulation to identify optimal decisions for resource allocation across multiple steps. The model evaluates profits by simulating all possible actions and selecting the one that maximizes expected returns.

- This approach ensures that decisions are not greedy but consider future implications of current actions.
- The state space exploration via dynamic programming helps discover globally optimal policies.
- Visualization through plots allows better understanding of profit evolution and decision dynamics.